

HW #3

①

1. ~~$\mathbf{IJK}$~~

$A(IJK)$: Inertial = Reference Frame

$B(ijk)$: Rotating Frame

$$\begin{aligned}\vec{\omega}(t) &= \vec{\omega}_{B/A}^B = \omega_x i + \omega_y j + \omega_z k \\ &= 0.1 i - 0.05 j + 0.03 k \text{ (rad/sec)} \\ &= \text{constant}\end{aligned}$$

$$\alpha(t) = \dot{\vec{\omega}} = 0$$

$$R \triangleq R_{B/A} \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = R \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

(a) i) From the attitude kinematics,

$$\dot{R} = \dot{S} R$$

$$\dot{R} = S(\vec{\omega}) R, \quad R(t=0) = I_{3 \times 3} \triangleq R_0$$

Since $\vec{\omega} = \text{constant}$,

$$R(t) = e^{S(\vec{\omega})t} R_0$$

$$\therefore R(t_f=10) = \begin{bmatrix} 0.8482 & -0.4606 & -0.2616 \\ 0.0140 & 0.5132 & -0.8582 \\ 0.5296 & 0.7242 & 0.4417 \end{bmatrix}$$

ii) ~~ϕ~~ From the given formulae,

$$\phi = \cos^{-1} \left[\frac{\text{trace}(R) - 1}{2} \right] = 1.1576 \text{ rad} = 66.3247 \text{ deg}$$

$$\vec{a} = \frac{1}{2 \sin \phi} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} = \begin{bmatrix} 0.8639 \\ -0.4319 \\ 0.2592 \end{bmatrix}$$

$$\Rightarrow \text{Exponential Coordinates } a\phi = \begin{bmatrix} 1 \\ -0.5 \\ 0.2 \end{bmatrix}$$

(2)

ii) Consider 3-2-1 rotation of Frame B w.r.t. Frame A by a sequence of ψ - θ - ϕ

$$R = R_{B/A} = R_1(\phi) R_2(\theta) R_3(\psi)$$

Solving nonlinear matrix eqn.

$$R_1(\phi) R_2(\theta) R_3(\psi) - R = 0 \quad (R \text{ is known from 1})$$

$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix} = \begin{bmatrix} 0.0171 & -0.4975 & 0.2647 \\ 0.5581 & 0.2647 & 1.0954 \\ 1.0228 & 1.0954 & 0.2647 \end{bmatrix} \text{ rad} = \begin{bmatrix} -28.5834 \\ 15.4264 \\ -62.7659 \end{bmatrix} \text{ deg}$$

N) From the def. of quaternion,

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} a_1 \sin \frac{\phi}{2} \\ a_2 \sin \frac{\phi}{2} \\ a_3 \sin \frac{\phi}{2} \end{bmatrix} = \begin{bmatrix} 0.4725 \\ -0.2363 \\ 0.1418 \end{bmatrix}$$

$$q_4 = \cos \frac{\phi}{2} = 0.8371$$

(b) Since $\vec{\omega} = \vec{\omega}_{B/A}^B = \text{constant}$ in Frame B,

$$\vec{\omega}_{B/A}^B(t=10) = \vec{\omega}_{B/A}^B(t=0) = \begin{bmatrix} 0.1 \\ -0.05 \\ 0.03 \end{bmatrix} \text{ rad/sec}$$

~~$$\vec{\omega}_{B/A}^A(t=10) = \vec{\omega}_{B/A}^A(t=0)$$~~

$$\vec{\omega}_{B/A}^A(t=10) = R_{A/B} \vec{\omega}_{B/A}^B(t=10)$$

$$= R_{B/A}^T \vec{\omega}_{B/A}^B(t=10)$$

$$= \begin{bmatrix} 0.1 \\ -0.05 \\ 0.03 \end{bmatrix}$$

What a coincidence!

$$2 \quad (I_x, I_y, I_z) = (2000, 2000, 1000) \text{ kg-m}^2$$

$$(M_x, M_y, M_z) = (0, 0, 0)$$

Z-axis is the axis of mass symmetry

(a) Based on the given information, the Euler's EoM is

$$I_x \dot{\omega}_x = (I_y - I_z) \omega_y \omega_z \quad \textcircled{1}$$

$$I_y \dot{\omega}_y = (I_z - I_x) \omega_z \omega_x \quad \textcircled{2}$$

$$I_z \dot{\omega}_z = 0 \quad \Rightarrow \omega_z = \text{const} \quad \textcircled{3}$$

$\therefore$ Equilibrium solution should satisfy

$$\begin{cases} \omega_y \omega_z = 0 \end{cases} \quad \textcircled{4}$$

$$\omega_z \omega_x = 0 \quad \textcircled{5}$$

i) $\omega_z = 0 \rightarrow \textcircled{4}, \textcircled{5} \text{ satisfied}$

$$\rightarrow \omega_B = \alpha i + \beta j + 0 \cdot k \text{ is an equilibrium}$$

ii) $\omega_z \neq 0 \rightarrow \omega_x = \omega_y = 0 \text{ for equilibrium}$

$$\rightarrow \omega_B = 0 \cdot i + 0 \cdot j + \omega_z k \text{ is equilibrium}$$

$\therefore$ All equilibrium solutions are

$$\vec{\omega} = \begin{bmatrix} \omega_x \\ \omega_y \\ 0 \end{bmatrix}$$

where ω_x and ω_y are arbitrary

$$\vec{\omega} = \begin{bmatrix} 0 \\ 0 \\ \omega_z \end{bmatrix}$$

where ω_z is arbitrary

④
(b) Since $\vec{\omega}_0(t=0) = \begin{bmatrix} 0 \\ 0 \\ 0.8 \end{bmatrix}$ lies on the eq. solution,

the angular velocity is constant, i.e.,

$$\vec{\omega}(t) \equiv \vec{\omega}(t=0) = \begin{bmatrix} 0 \\ 0 \\ 0.8 \end{bmatrix}$$

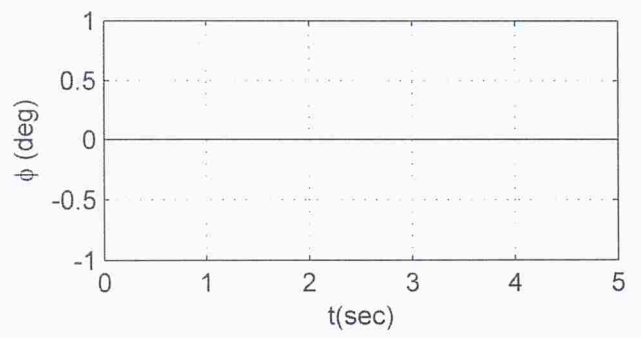
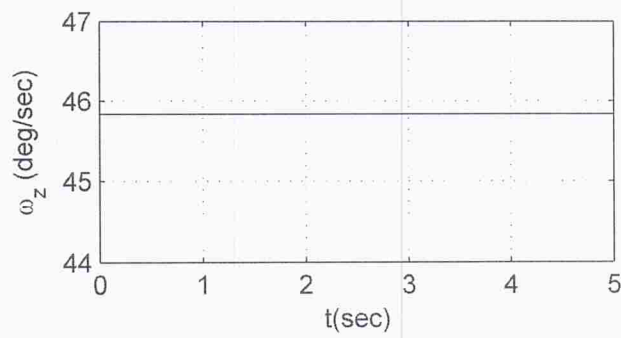
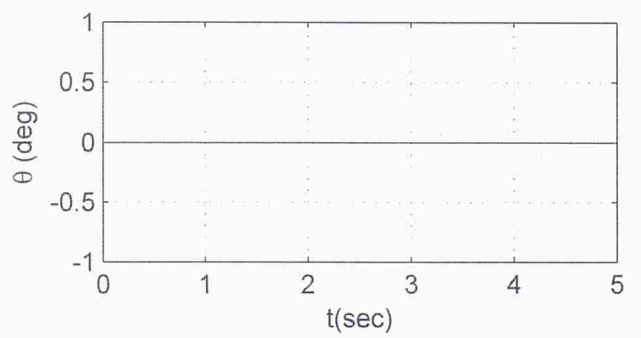
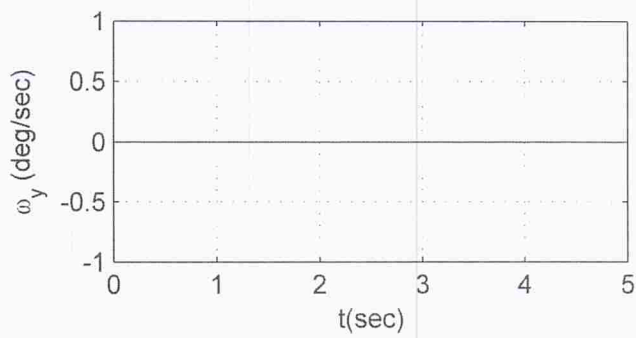
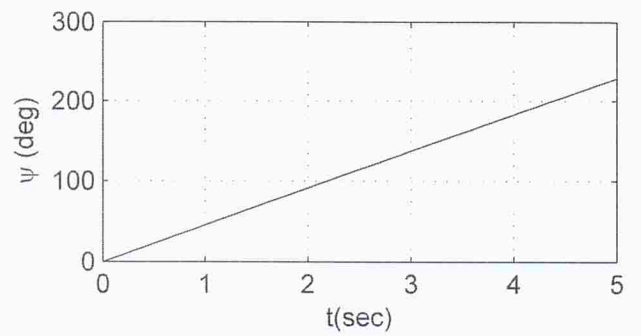
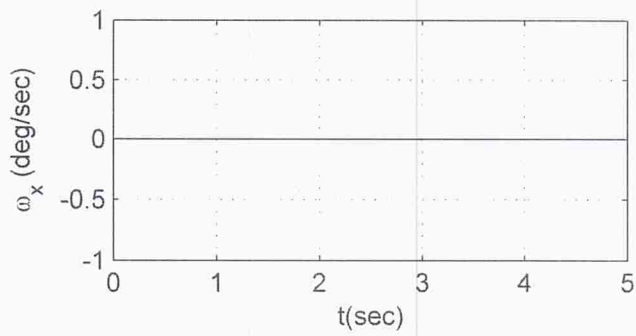
Also from the ~~attitude~~ Euler angle attitude dynamics,

$$\begin{cases} \dot{\psi} = (\cancel{\omega_y \sin \phi} + \omega_z \cos \phi) \sec \theta, & \psi(0) = 0 \\ \dot{\theta} = \cancel{\omega_y \cos \phi} - \omega_z \sin \phi, & \theta(0) = 0 \\ \dot{\phi} = \cancel{\omega_x} + \cancel{\omega_y \sin \phi \tan \theta} + \omega_z \cos \phi \tan \theta, & \phi(0) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \dot{\psi}(t) = \omega_z = \text{const} \\ \dot{\theta}(t) = 0 = " \\ \dot{\phi}(t) = 0 = " \end{cases}$$

$$\Rightarrow \begin{cases} \psi(t) = \omega_z t \\ \theta(t) = 0 \\ \phi(t) = 0 \end{cases}$$

which justifies the numerical responses on the next page



(C) Since $\vec{\omega}(t=0) = \begin{bmatrix} 0.05 \\ -0.05 \\ 0 \end{bmatrix}$ lies on the equilibrium,

the angular velocity is constant, i.e.,

$$\vec{\omega}(t) = \vec{\omega}(t=0) = \begin{bmatrix} 0.05 \\ -0.05 \\ 0 \end{bmatrix}$$

Also, from the attitude dynamics
Euler-Angle

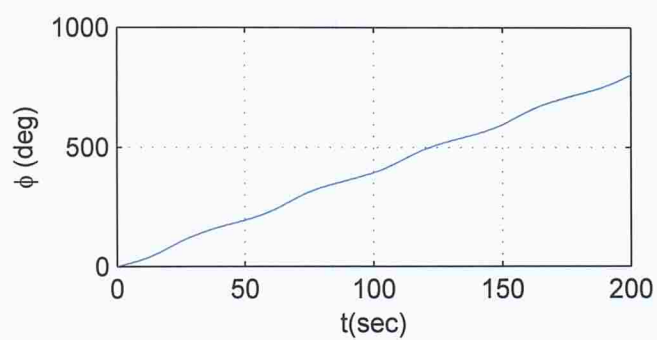
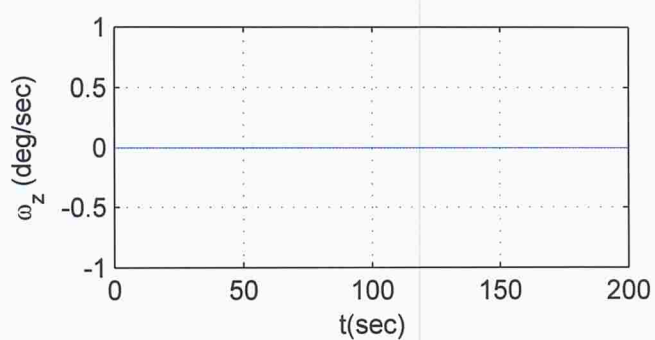
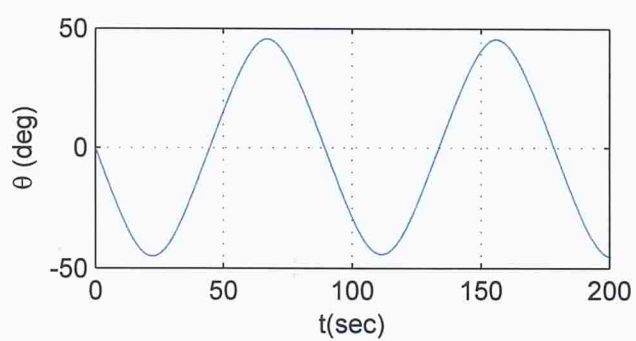
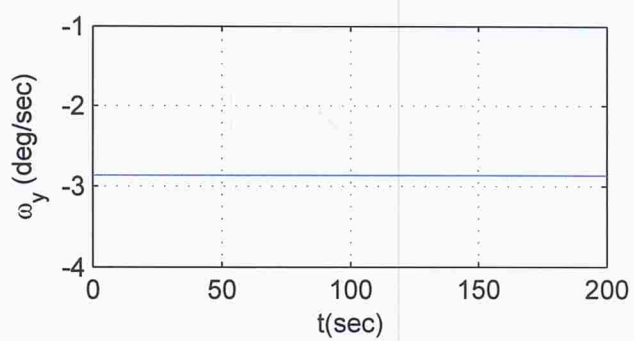
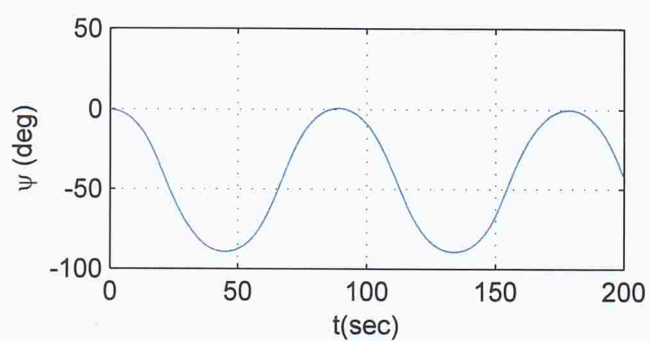
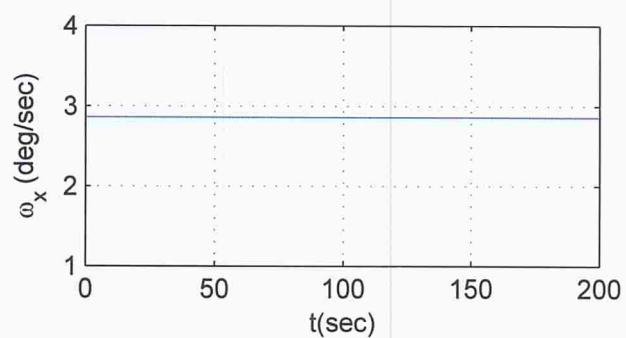
$$\begin{cases} \dot{\psi} = \omega_y \sin \phi \sec \theta, & \psi(0) = 0 \\ \dot{\theta} = \omega_y \cos \phi, & \theta(0) = 0 \\ \dot{\phi} = \omega_x + \omega_y \sin \phi \tan \theta, & \phi(0) = 0 \end{cases}$$

Numerical responses are provided on the next page.

The period of $\psi(t)$ is $T_\psi \approx \frac{90}{0.05} \text{ sec}$
 " $\theta(t)$ is $T_\theta \approx 90 \text{ sec}$

7

Figure 51



3. $(I_x, I_y, I_z) = (2000, 1500, 1000) \text{ kg-m}^2$

$$(M_x, M_y, M_z) = (0, 0, 0)$$

(a) Similarly as in Prob 2.

$$I_x \omega_x = (I_y - I_z) \omega_y \omega_z$$

$$I_y \omega_y = (I_z - I_x) \omega_z \omega_x$$

$$I_z \omega_z = (I_x - I_y) \omega_x \omega_y$$

Equilibrium should satisfy

$$\begin{cases} \omega_y \omega_z = 0 \\ \omega_z \omega_x = 0 \\ \omega_x \omega_y = 0 \end{cases}$$

All eq. solutions are

$$\omega = \begin{bmatrix} 0 \\ 0 \\ \omega_z \end{bmatrix}$$

$$\omega = \begin{bmatrix} 0 \\ \omega_y \\ 0 \end{bmatrix}$$

$$\omega = \begin{bmatrix} \omega_x \\ 0 \\ 0 \end{bmatrix}$$

where ω_x, ω_y , and ω_z are arbitrary

(b) Since $\vec{\omega}(t=0) = \begin{bmatrix} 0 \\ 0 \\ 0.8 \end{bmatrix}$ lies on the equilibrium,

the ang. velocity is constant, i.e.
 $\vec{\omega}(t) \equiv \vec{\omega}(t=0) = \begin{bmatrix} 0 \\ 0 \\ 0.8 \end{bmatrix}$

the angular velocity is constant, i.e.

$$\vec{\omega}(t) = \vec{\omega}(t=0) = \begin{bmatrix} 0 \\ 0 \\ 0.8 \end{bmatrix}$$

Also from the Euler-Angle attitude dynamics,

$$\begin{cases} \dot{\psi} = \omega_z \cos \phi \sec \theta, & \psi(0) = 0 \\ \dot{\theta} = -\omega_z \sin \phi, & \theta(0) = 0 \\ \dot{\phi} = \omega_z \cos \phi \tan \theta, & \phi(0) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \dot{\psi} = \omega_z \\ \dot{\theta} = 0 \\ \dot{\phi} = 0 \end{cases}$$

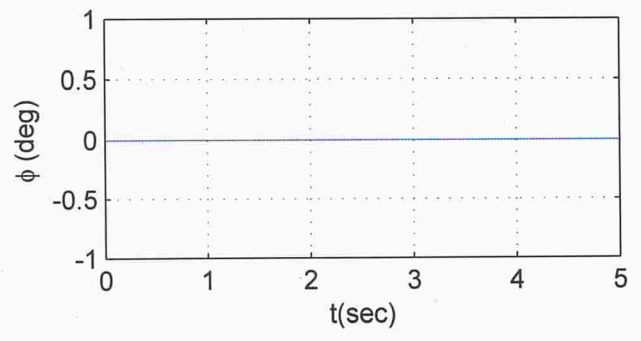
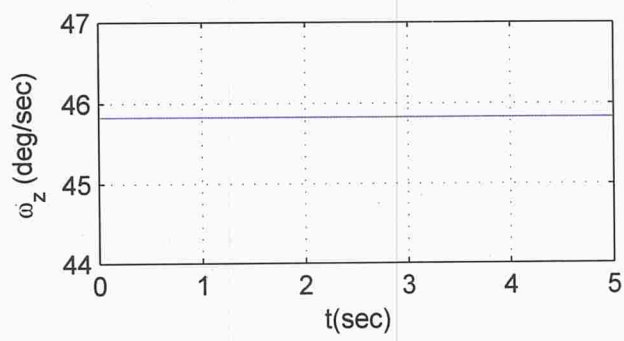
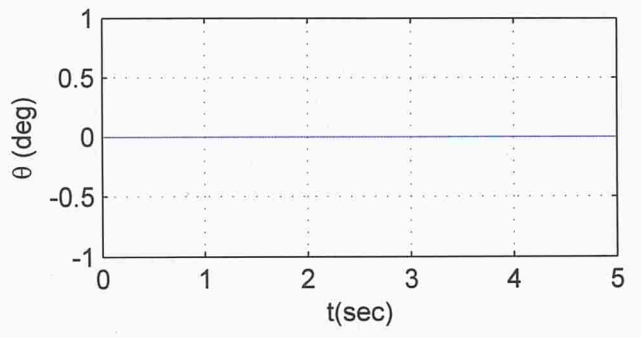
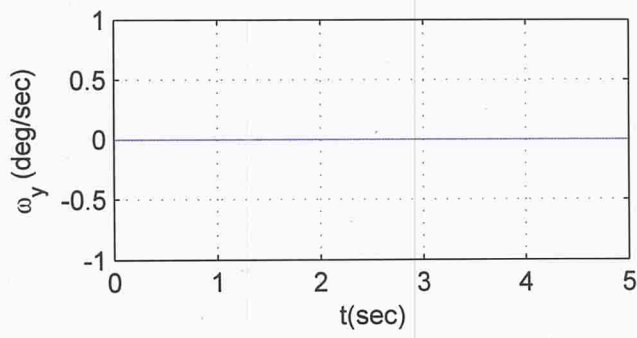
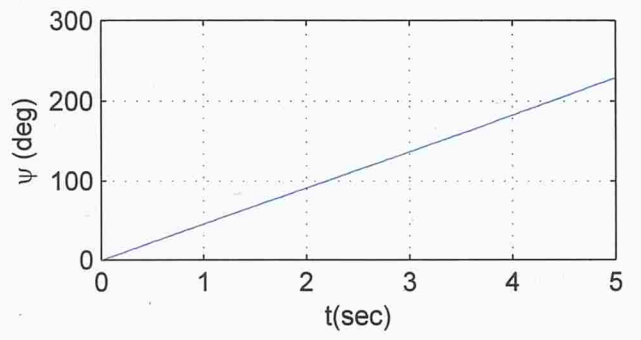
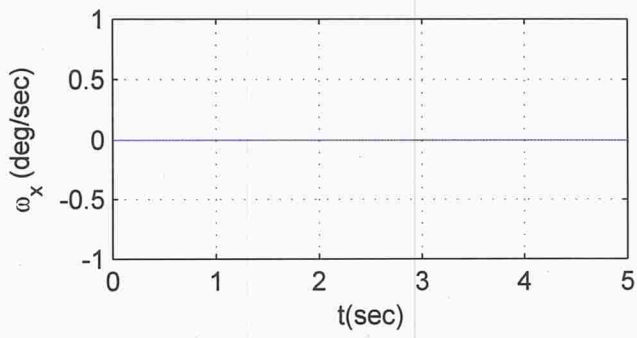
$$\Rightarrow \begin{cases} \psi(t) = \omega_z t \\ \theta(t) = 0 \\ \phi(t) = 0 \end{cases}$$

which justifies the numerical responses on the next page.

(c) Numerical responses show ~~that~~ oscillatory behaviors:

$$\begin{cases} T_{wx} = 1 \text{ wsec} \\ T_{wy} = 2 \text{ wsec} \\ T_{wz} = 2 \text{ wsec} \end{cases}$$

$$\begin{cases} T_{\psi} \approx 185 \text{ sec} \\ T_{\theta} \approx 185 \text{ sec} \end{cases}$$



15

